Green Chemistry

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Cite this: *Green Chem.*, 2023, **25**, 4808

Sustainable polar aprotic/poly-deep eutectic solvent systems for highly efficient dissolution of lignin†

Qiaoling Liu,‡[a] Yang Wang,‡[a] Jing Bian,[a] Ming-Fei Li,[a] Jun-Li Ren,[b] Xiang Hao *[a] and Feng Peng *[a]

Received 3rd March 2023,
Accepted 22nd May 2023

DOI: 10.1039/d3gc00733b

rsc.li/greenchem

Developing green and effective solvents is of significant importance for the utilization of lignin. Here, we demonstrated that biomass-derived polar aprotic solvents can serve as co-solvents with poly-deep eutectic solvents (PDESs) for massively dissolving various lignin samples at room temperature. The pristine PDESs were composed of polyethylene glycol derivatives and levulinic acid, while all of them exhibited poor solubility to enzymatic hydrolysis lignin (EHL), sodium lignosulfonate (LS), and organosolv lignin (OL). With the further introduction of polar aprotic solvents, including dimethylisosorbide (DMI), dihydrolevoglucosenone (Cyrene), *N*-formylmorpholine (NFM), and propylene carbonate (PC), synergy between the polar aprotic solvents and PDESs dramatically increased lignin solubility to 96.82%. PC and DMI were more suitable for assisting the solubilization of EHL and OL, while Cyrene and PC were targeted at the solubilization of LS in PDESs, respectively. Moreover, a new unified co-solvent solubilization mechanism relative to the Kamlet–Taft empirical parameters was proposed. The novel polar aprotic/PDES systems provide a new avenue for boosting the high utilization of lignin.

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Introduction

Lignin, a three-dimensional (3D) amorphous aromatic compound, is considered the second most abundant biopolymer.[1] Owing to its environmentally friendly, large-scale availability, and renewable properties, the application of lignin has attracted fast-growing attention.[2,3] In general, more than 100 million tons of lignin are produced annually in pulp and paper processes, yet the large majority of lignin is burned to generate energy or discharged into waterways.[4] The effective use of lignin provides a breakthrough pathway for its high-value utilization at scale. However, the structures of lignin are complex and irregular reticular due to the preparation methods and the origin of the resource.[5,6] Thus, lignin solubility in most common solvents is generally not satisfactory.

To date, much effort has been devoted to the dissolution of lignin. Several solvent systems (*e.g.*, dilute acid, alkali, ammonia, and ethanol) can dissolve lignin successfully, yet their applications suffer from the volatility and toxicity of organic solvents.[7–9] As a green alternative, ionic liquids (ILs) have received widespread attention and have been extensively exploited for biomass dissolution. However, their applications are limited by certain drawbacks, such as inherited toxicity, poor biodegradability, and high cost.[10–12] Deep eutectic solvents (DESs), known as IL analogues, are synthesized by simply mixing the hydrogen-bond acceptor (HBA) and hydrogen-bond donor (HBD) at a certain temperature, avoiding the use of toxic and expensive solvents.[13–15] Nonetheless, the common DESs are focused on small-molecular-based complexes, while DESs involving polymer have little been researched. Polyethylene glycol with low cost, easy degradability, and chemical stability is regarded as a green polymer.[16] A few groups have reported that polyethylene glycol with flexible alkoxy groups, such as HBA, could react with the hydrogen of hydroxyl groups in the cellulose, significantly promoting the dissolution of cellulose.[17,18] Zang *et al.*[19] further showed that a near-neutral DES (composed of choline chloride, boric acid, and polyethylene glycol-200) could separate lignin, hemicelluloses, and retain cellulose when treating wheat straw at 120 °C for 4 h. More recently, poly-deep eutectic solvents (PDESs) composed of polyethylene glycol derivatives, *e.g.*, polyethylene glycol 200, 1000, poly(ethylene glycol) dimethyl ether as HBAs, carboxylic acids and amides as HBDs, were proposed by Mu *et al.*[16,20,21] The resultant solvents could be successfully applied to dissolve both refractory metal oxides and transition metal oxides,[20] which provided a new insight toward the potential application of PDESs. However, it is worth noting that these

[a]*Beijing Key Laboratory of Lignocellulosic Chemistry, MOE Engineering Research Center of Forestry Biomass Materials and Energy, Beijing Forestry University, Beijing 100083, China. E-mail: fengpeng@bjfu.edu.cn*
[b]*State Key Laboratory of Pulp and Paper Engineering, South China University of Technology, Guangzhou 510640, China*

†Electronic supplementary information (ESI) available. See DOI: https://doi.org/10.1039/d3gc00733b

‡These authors contributed equally to this work.

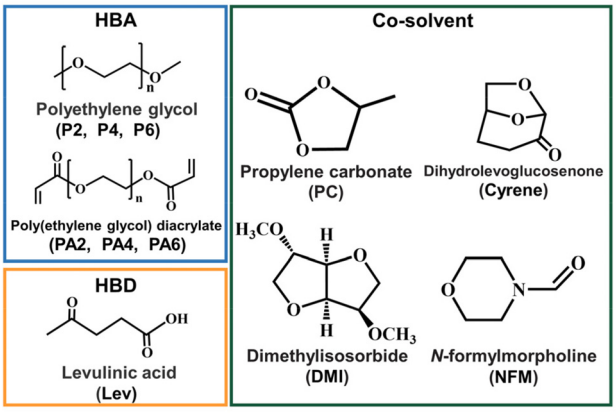
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PDESs are usually very viscous and poorly flowable, resulting in the unsatisfactory solubilization of many polymers. Therefore, these PDESs are rarely used in lignin dissolution.

Apart from the aforementioned solvent systems, polar aprotic solvents with unique characteristics (*e.g.*, solubilizing properties, high polarity, low boiling point, and low reactivity) have attracted much attention in a variety of areas.[22–24] Nevertheless, the traditional polar aprotic solvents extracted from fossil lead to enormous energy consumption, environmental pollution, and present a serious hazard to human health.[21,23] Recently, several polar aprotic organic solvents derived from biomass have been successfully developed. For example, dihydrolevoglucosenone (Cyrene) derived from cellulose is a potential substitute for the widely used *N*,*N*-dimethylformamide (DMF). Avérous *et al.*[6] identified that Cyrene/H_2O formed a geminal diol structure that could efficiently dissolve various lignin samples. Alternatively, dimethylisosorbide (DMI), a nontoxic, high-boiling green solvent stemming from D-sorbitol, is considered one of the top-10 bio-based platform chemicals.[25] Wang and co-works indicated that DMI/H_2O can serve as a co-solvent for the treatment of *Eucalyptus* assisted by 72 mM H_2SO_4 at 120 °C for 60 min, in which most of the lignin and hemicelluloses could be effectively removed, and the majority of the retained cellulose was near-completely hydrolysis.[26] These works prompted us to determine whether these solvents could be combined with PDESs for effectively dissolving lignin at room temperature.

Herein, a series of polar aprotic/PDESs with great biocompatibility, low cost, and low toxicity were designed by simply mixing bio-based levulinic acid (Lev) and polyethylene glycol (denoted as P-based PDESs), and poly(ethylene glycol) diacrylate (denoted as PA-based PDESs) with different molecular weights (M_w = 200, 400, and 600 Da) at room temperature, employing green multifunctional polar aprotic solvents as co-solvents derived from lignocellulose (DMI and Cyrene), non-reprotoxic *N*-formylmorpholine (NFM), and propylene carbonate (PC) (Scheme 1). We found that the mixed co-solvent/PDESs had great performances in dissolving various types of lignin, including enzymatic hydrolysis lignin (EHL), organosolv lignin (OL), and sodium lignosulfonate (LS) at room temperature. The solubilization behavior of lignin in the co-solvents/PDESs was further analyzed, and a unified dissolution mechanism applicable to different lignin samples was proposed.



Scheme 1 Chemical structures of HBA, HBD, and the co-solvents. P2, P4, and P6 represent polyethylene glycol with M_w 200, 400, and 600 Da, respectively. PA2, PA4, and PA6 represent poly(ethylene glycol) diacrylate with M_w 200, 400, and 600 Da, respectively.

Results and discussion

Dissolution of lignin in different solvent systems

The green PDESs were prepared by simply mixing HBA and HBD (Scheme 1) with a molar ratio of 1 : 2 at room temperature for 1 h, and the detailed procedure is described in the Experimental section. For practical applications, we determined the phase-transition temperatures (Fig. S1†), viscosity, and density of the PDESs (Table S1†). It should be pointed out that the viscosity of the PDESs decreased with M_w increasing from 200 to 600.

Green polar aprotic solvent/PDESs were fabricated by mixing six PDESs (P2-Lev, P4-Lev, P6-Lev, PA2-Lev, PA4-Lev, and PA6-Lev) and four polar aprotic solvents (NFM, DMI, PC, and Cyrene) in varying volume ratios. The solubilities of lignin (EHL, OL, and LS) were further determined at room temperature in the prepared solvent systems (Fig. 1–3 and Tables S4–S6†). Compared to the solubility of EHL in pure P2-Lev or co-solvent, the solubilities of EHL increased to 80.23%, 73.17%, and 72.17% in P2-Lev with 20 vol% DMI, NFM, and PC, respectively (Fig. 1a and Tables S2, S4†), implying that P2-Lev and polar aprotic solvents (NFM, DMI, and PC) worked synergistically. On the contrary, the solubility of EHL in P2-Lev decreased after adding Cyrene compared to that in pure P2-Lev, indicating that Cyrene may reduce the solvating capability of the P2-Lev to act as an anti-solvent. With the increase in co-solvent content (20–80 vol%), the solubilization effect of the co-solvent on the dissolution of EHL decreased significantly, and the maximum solubility value of EHL occurred at 20 vol% of polar aprotic solvents, allowing access to fractionating the

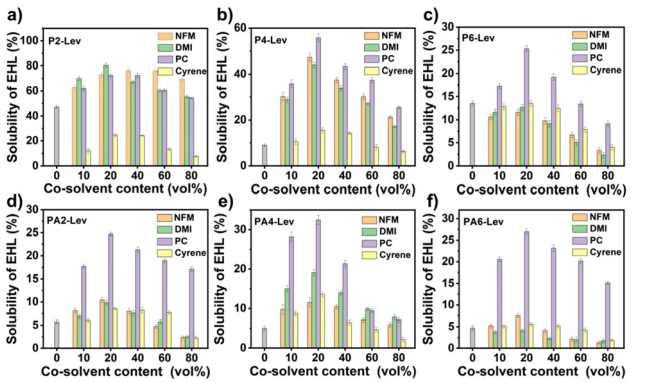


Fig. 1 Solubility of EHL at room temperature in mixtures of (a) P2-Lev, (b) P4-Lev, (c) P6-Lev, (d) P2-Lev, (e) P4-Lev, (f) P6-Lev and four polar aprotic solvents (NFM, DMI, PC, and Cyrene).

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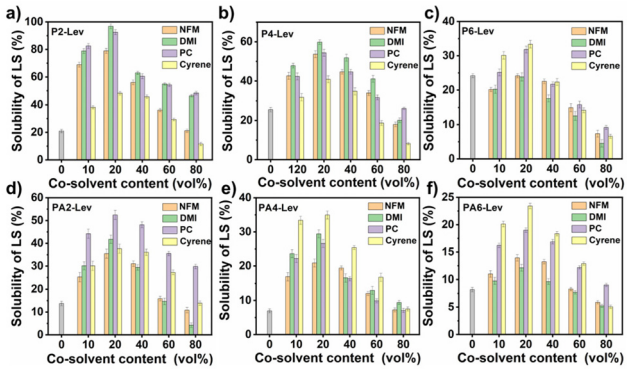


Fig. 2 Solubility of LS at room temperature in mixtures of (a) P2-Lev (b) P4-Lev (c) P6-Lev (d) P2-Lev (e) P4-Lev (f) P6-Lev and four polar aprotic solvents (NFM, DMI, PC, and Cyrene).

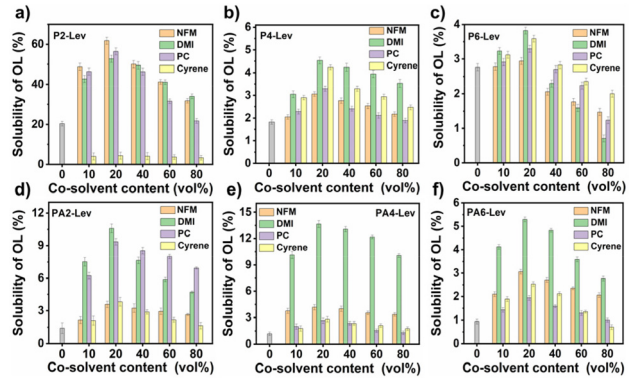


Fig. 3 Solubility of OL at room temperature in mixtures of (a) P2-Lev (b) P4-Lev (c) P6-Lev (d) P2-Lev (e) P4-Lev (f) P6-Lev and four polar aprotic solvents (NFM, DMI, PC, and Cyrene).

solubilized lignin by diluting the solution directly. When the M_w of the P-based PDESs was increased from 200 to 400, the EHL solubility decreased significantly (Fig. 1b and Table S2,† entry 2), yet all the polar aprotic solvents (NFM, DMI, PC, and Cyrene) in P4-Lev effectively promoted the dissolution of EHL at room temperature, regardless of the co-solvent volume ratio (Fig. 1b and Table S4†). However, when the M_w of P-based PDESs was increased to 600, the solubility of EHL slightly increased with the increase in the M_w (Fig. 1c and Table S2,† entry 3), indicating that there was no obvious correlation between the solubility of EHL and the degree of polymerization of the PDESs. Such a phenomenon differed from the case in which the lignin solubility was enhanced in the PDESs with increasing the carbon chain length.[27] Besides, only PC as a co-solvent in four polar aprotic solvents contributed to the dissolution of EHL in P6-Lev with the best solubility of 25.29% (Fig. 1c and Table S4,† entry 9).

The solubility of EHL changed slightly in the PA-based PDESs, which was worse than that in the P-based PDESs (Fig. 1d–f and Table S2†). We speculated that the additional aldehyde group increased the steric hindrance between the PDESs and lignin interaction. When PA-based PDESs with lower M_w (PA2-Lev and PA4-Lev) were combined with polar aprotic solvents, all the polar aprotic solvents showed an enhanced solubilization of EHL (Fig. 1d, e and Table S4†). However, for PA6-Lev, only PC serving as co-solvent effectively solubilized EHL (Fig. 1f and Table S4†). In general, the P-based PDESs showed better performance in the EHL solubility than the PA-based PDESs. Meanwhile, the solubilization effect of EHL depended on the type and amount of PDESs and polar aprotic solvents. The solubilization effect of EHL significantly decreased when the content of co-solvents increased from 20 to 80 vol% and all the polar aprotic solvents (except 20 vol% Cyrene/P2-Lev) mixed with a lower M_w of PDESs showed an outstanding solubilization effect for EHL, yet when the M_w of the PDESs was increased, only PC contributed to the enhanced solubility of EHL.

More lignin solubility tests were performed to screen the solvating capacity of PDESs mixed with different polar aprotic solvents, and the dissolution behavior of LS was determined (Fig. 2 and Table S5†). First, the dissolution of LS in different PDESs were measured. Similar to the dissolution behavior of EHL in PDESs, there was no obvious correlation between the LS solubility and the M_w of PEG in the PDESs. P4-Lev provided the best performance in the dissolution of LS (25.52%) in the P-based PDESs at room temperature (Fig. S2 and Table S2,† entry 1). The solubility of LS in the PA-based PDESs decreased with changing PA2 to PA4, but increased for the PA6-based PDESs. PA2-Lev was one of the best PA-based PDES, with a solubility of 13.70% at room temperature for LS (Fig. S2 and Table S2,† entry 5).

Synergy was also observed between P2-Lev and the polar aprotic solvents. All the compounds could solubilize LS at room temperature after adding polar aprotic solvents to P2-Lev, and the solubilities of LS in 20 vol% NFM/DMI/PC increased from 20.94% to 78.94%, 96.82%, and 92.52%, respectively (Fig. 2a and Table S5,† entry 2). When the M_w of the P-based PDESs was changed from 200 to 400, the solubility of LS roughly decreased, but all the polar aprotic solvents still promoted the dissolution of LS at room temperature effectively, in which NFM, PC, and DMI played an obvious role. When the M_w of the P-based PDESs was increased to 600, Cyrene and PC dramatically improved the dissolution of LS. Similarly, the PA-based PDES mixed with polar aprotic solvents showed irreplaceable synergy in dissolving LS (Fig. 2d–f). PC was one of the best polar aprotic solvents as a co-solvent in PA2-Lev, with a great solubility of 52.47% (Fig. 2d and Table S2 (entry 4), Table S5, (entry 17)†). However, when the M_w was varied from 200 to 400/600, the best choice of co-solvent was Cyrene (Fig. 2e and f). All the polar aprotic solvents/PDESs could improve the solubility of LS except for NFM and DMI in P6-Lev, in which PC was the best one for the dissolution of LS.

Generally, OL was the most difficult to dissolve among the three examined types of lignin. Indeed, OL was not sufficiently soluble in the prepared PDESs in most cases (Fig. 3 and Table S6†). Similarly, a maximum value of only 20. 41% could be achieved in P2-Lev (Fig. 3a and Table S2,† entry 1). The

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solubilities of OL were negligible in P4-Lev and P6-Lev, suggesting that a lower M_w was beneficial for the dissolution of OL, resulting from their lower viscosity and greater opportunity for hydrogen-bond formation with more HBA (Table S1†).[28] In addition, the P-based PDESs showed better performance than the PA-based PDESs in the dissolution of OL. With the incorporation of co-solvents, NFM, DMI, and PC showed an excellent ability to promote lignin dissolution in P2-Lev (Fig. 3a and Table S6†). Since the optimal solvent efficiency for dissolving lignin occurs at lower co-solvent concentrations, the solubility of OL in 20 vol% NFM/P2-Lev increased from 20.41% to 61.76% (Fig. 3a and Table S6,† entry 2), and the solubilization effect was weakened when the M_w of the P-based PDESs increased (Fig. 3a–c). With regard to the P4- and P6-based PDESs, all the polar aprotic solvents, especially DMI and Cyrene, showed an enhanced solubilization effect for OL. All the polar aprotic solvents contributed to the dissolution of OL, while only DMI and PC showed a dramatically increased dissolution ability for OL in the PA-based PDESs. Specifically, PC was the preferred co-solvent for efficient dissolving OL in PA2-Lev, while DMI was still dominant for the dissolution of OL in all the PA-based PDESs (Fig. 3d–f).

Here, the solubilities of the three lignin samples (EHL, LS, and OL) in the P-based PDESs were superior to the PA-based PDESs. For EHL, NFM, PC, and DMI promoted the solubility of lignin, in which PC exhibited a better solubilization in higher M_w PDESs, especially in the PA-based PDESs. For LS, synergy between the polar aprotic solvents and PDESs was observed (except for NFM, DMI in P6-Lev). Cyrene was preferred to the solubilization in the PDESs with a higher M_w, while PC was more suitable in lower M_w PDESs. For OL, all the polar aprotic solvents (except Cyrene/P2-Lev) played an obvious role in the solubilization effect for OL; in particular, the strong solvating capacity of DMI as a co-solvent made it more efficient in dissolving OL in all the PDESs.

For the correlation between the chemical structure of the different lignin samples and their solubilities, two-dimensional heteronuclear single quantum coherence (2D HSQC) NMR spectra of the three lignin samples were obtained to reveal their corresponding linkage structures and monomer compositions (Fig. S3†), and the signals were assigned according to previously reported results (Table S7†).[29] The presence of apparent signals for methoxy groups (OMe) in the side-chain region indicated that these lignin samples contained abundant guaiacyl-type (G) and syringyl-type (S) units. The pronounced cross-signals corresponding to substructures of β-O-4 linkages (A) for side chains were found in EHL and OL; however, the C–H signals at the A_α and A_β positions were not detected in the spectrum of LS. Similarly, the characteristic signal corresponding to C_γ–H_γ of the phenylcoumarin substructure (C) was obviously weakened in LS. These results indicated that the linkage structures of LS were significantly cleaved during treatment. Thus, the difficulty of dissolution was decreased, causing the relatively higher solubility of LS in the co-solvent/PDESs. In the aromatic region, signals for the S, G, and *p*-hydroxyphenyl (H) units were detected in the three lignin samples, indicating that these lignin samples were derived from herbaceous plants. The obvious condensation signal of condensed S units was detected at around 103.5–106.5/6.15–6.45 ppm in the spectrum of OL, which might greatly limit the solubility of OL. In the aromatic region of LS, the chemical shifts of the G and H units changed significantly, indicating that the presence of the sulfonate groups led to a reduction in the electron cloud density of the aromatic rings. Meanwhile, the contained sulfonate groups afforded LS with much higher polarity, thus leading to the special dissolution behavior of LS.

Rheological characteristics of DMI/P2-Lev solutions with different EHL concentrations

The rheological properties of polymer solutions are important for exploring the nano-microstructure of lignin macromolecules in co-solvent/PDES solutions. The apparent viscosity of DMI/P2-Lev solutions with different EHL concentrations as a function of temperature are presented in Fig. 4a, providing the molecular conformation, overlap, and entanglement concentration.[30] DMI/P2-Lev displayed a relatively lower η value (47.18 mPa·s at 20 °C) than the common ILs (*e.g.*, [AMIM][Cl], [C_4mim][BF_4], and [DBUH][$CH_3CH_2OCH_2COO$]) and DESs (*e.g.*, K_2CO_3-glycerol, ChCl-urea, and ChCl-glycerol) reported in previous literature,[31–34] which was beneficial to the mass transfer of lignin between DMI/P2-Lev, and improved the solubility of lignin.[35,36] By increasing the concentration of EHL from 1.0% to 20% at 20 °C, the η value substantially was increased from 49.90 to 89.34 mPa·s due to the entanglement of the macromolecular chains, which is a typical rheological property of macromolecular solutions.[30] Conversely, the η values of DMI/P2-Lev solutions with different EHL concentrations dropped with increasing the temperature from 20 °C to 80 °C at a given shear rate (*e.g.*, 9.61 mPa·s at 80 °C for the 20% solution). This could be ascribed to the fact that the lignin macromolecular network structure will decompose after raising the temperature, and the thermal motion of macromolecules can be intensified.[31]

The flow curves in terms of the η and shear rate relationship for DMI/P2-Lev solutions with different EHL concentrations at 20 °C are presented in Fig. 4b. The η of EHL solution decreased with the increase in shear rate, exhibiting a pro-

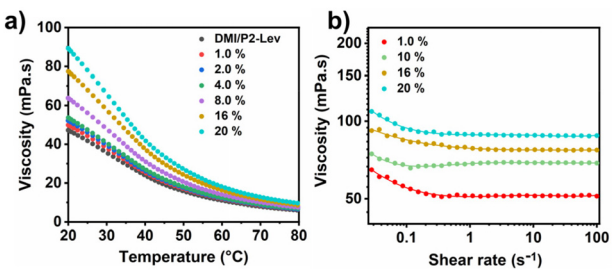


Fig. 4 Rheological behavior of 20 vol% DMI/P2-Lev solutions with different EHL concentrations. (a) viscosity (η) curves as a function of (a) temperature and (b) shear rate at 20 °C.

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minent shear-thinning behavior within the tested shear rate range (<1 s^{-1}). This was because the entanglement network structure between EHL chains was destroyed with the increase in shear rate.[37] Thus, the corresponding shear-thinning behavior was observed. When the shear rate was ramped up above 1 s^{-1}, the flow curve of EHL exhibited the characteristics of a Newtonian fluid as the viscosity remained almost constant with respect to the shear rate, indicating that the EHL structure dissolved in DMI/P2-Lev became fragile and could easily be disturbed at low shear rates, aligned in the direction of the shear.[30,37]

Dissolution mechanism of various types of lignin in the polar aprotic/PDES systems

The interactions between the polymer and the solvent molecules play a crucial role in the determination of lignin's solubility.[38] Generally, these interactions can be estimated by the Kamlet–Taft empirical parameters (*e.g.*, α, β, π^*) of solvents, which quantifies the hydrogen-bond-donating ability (acidity), hydrogen-bond-accepting ability (basicity), and polarity of the solvent.[39–41] To better explore the dissolution mechanism of lignin in the PDESs as well as the role of the co-solvents, the Kamlet–Taft empirical parameters of pure PDESs, polar aprotic solvents, and co-solvent/PDESs were determined (Tables S7–S11†). First, the correlation between the efficiency of the PDESs used to dissolve lignin and the Kamlet–Taft empirical parameters was interpreted. The Kamlet–Taft empirical parameters of the PDESs varied depending on the HBA structure (Fig. 5a–c). There was no correlation between the β value of the PDESs and the M_w of the corresponding solvents. The π^* value decreased with increasing the M_w of HBA, indicating that a higher degree of polymerization of the PDESs resulted in a decreased polarizability of the PDESs, and the charge between the ions was highly delocalized.[42] Meanwhile, the α value of the PDESs decreased with increasing the M_w of HBA, which may be due to the OH group in P6 not being as strong as the P2 in forming hydrogen bonds with Lev.[42]

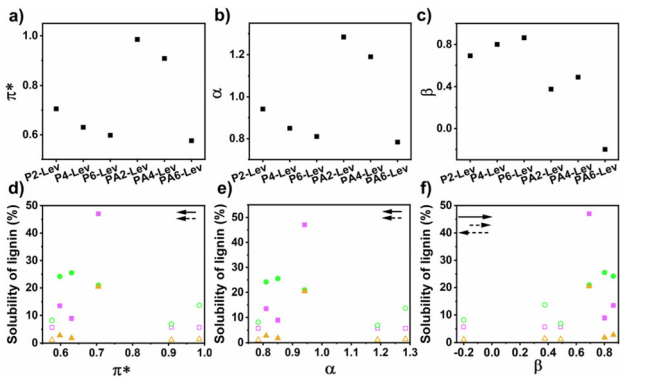


Fig. 5 Kamlet–Taft empirical parameters of the PDESs (a–c). Correlation between the lignin solubility and the Kamlet–Taft empirical parameters of the PDESs (d–f). Solid and hollow indicate the solubility of lignin in the P-based PDESs and PA-based PDESs, respectively. The shapes of the plots correspond to the solubility of lignin as follows: square: EHL, circle: LS, and triangle: OL. The direction of the arrow indicates the Kamlet–Taft empirical parameters of the PDESs from low to high M_w.

The solubilities of EHL and OL first decreased and then increased with increasing the M_w of the P-based PDESs (Fig. 1, 3 and Tables S4, S6†), and P2-Lev showed the best performance in dissolving lignin compared with the other PDESs due to the higher π^* (Table S6†). It could be estimated that the stronger solvating capability of P2-Lev interacted with the aromatic parts of lignin through dispersion interactions.[38] P6-Lev demonstrated a superior dissolution performance for EHL and OL than P4-Lev, because the higher β value contributed to breaking the intermolecular hydrogen bonds of lignin (Table S6†).[38] Thus, hydrogen bonding played a very important role in the dissolution of lignin more than the dispersion interaction in this case. For LS (Table S5† and Fig. 3), its dissolution behavior was different from those of EHL and OL. The solubility first increased and then decreased with increasing the M_w in the P-based PDESs, wherein P4-Lev with a higher β value showed the best performance (Table S6†). The solubility decreased with the M_w increasing from 400 to 600 due to the decrease in π^* (Table S6†). The dissolution behavior of LS in the PA-based PDESs was different, with the solubility tending to decrease and then increase with increasing the M_w, which corresponded to increasing the π^* and β values, respectively.

From the above results, we could conclude that lignin dissolution in PDESs is complex, and the type of lignin, and the polarity and hydrogen-bond-accepting ability of the solvent all play important roles in exploring the dissolution mechanism of lignin. We see some stark differences in the Kamlet–Taft empirical parameters (*e.g.*, α, β, π^*) of the solvents when polar aprotic solvents participated in the prepared PDESs. An increase in β may be brought about by disrupting the original hydrogen-bonding network of PDESs, giving rise to more oxygen atoms of the alkoxy group (CH_2–CH_2–O).[19,43,44] The α and π^* values of PDESs roughly decreased when adding the four different co-solvents. Unfortunately, we did not find a correlation between the α and β values and lignin solubility (Fig. S3–S8†), yet the π^* value of the co-solvent/PDESs was related to the solubility of lignin, which may be due to the extended π-electron network of lignin, making dispersion interactions more important for the dissolution of lignin (Fig. 6 and Fig. S9, S10†).

The correlation between the lignin solubility and the Kamlet–Taft parameters of the corresponding solvents was initially evaluated using LS solubility as a model (Fig. 6). The π^* parameter was measured for evaluation of the dispersion forces of the solvent and the aromatic parts of the polymer.[38] It was observed that the co-solvent/PDES with π^* exceeding a critical value was able to dissolve LS at room temperature. For example, the co-solvent exhibited synergy with P2-Lev when $\pi^* > 0.441$ for the solvent, yet the LS solubility in 80 vol% Cyrene/P2-Lev was poorer than that in pure P2-Lev (Fig. 6a), which may be due to the lower α value of 80 vol% Cyrene/P2-Lev (Table S11†), resulting in a weak hydrogen-bonding

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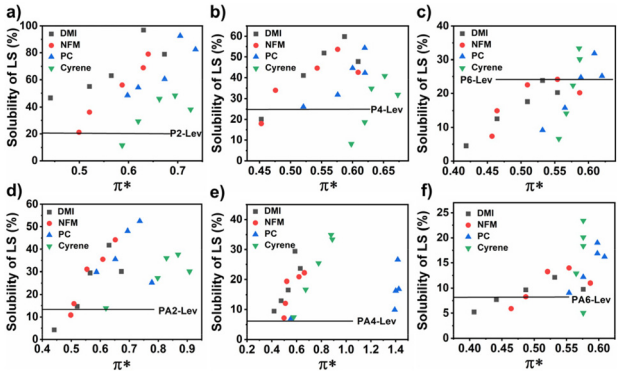


Fig. 6 Correlation between the LS solubility and the π* value of the corresponding solvent (the line represents the solubility of the lignin in PDES, and the data above the line indicate that lignin was promoted to dissolve by adding co-solvents, and the critical values of π* were 0.441, 0.475, 0.588, 0.509, 0.418, and 0.486 in (a) P2-Lev, (b) P4-Lev, (c) P6-Lev, (d) PA2-Lev, (e) PA4-Lev, and (f) PA6-Lev, respectively).

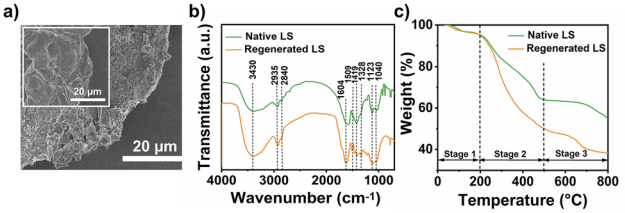


Fig. 7 Characterization of native and regenerated lignin. (a) SEM images of regenerated LS (inset showing the morphology of the initial LS) (b). FT-IR spectra of native and regenerated LS. (c) TGA curves of native and regenerated LS.

capacity formation with the lignin.[45] Similarly, LS could be efficiently solubilized in P4-Lev after adding various co-solvents with a π* value of co-solvent (except for 60%–80% Cyrene)/P4-Lev > 0.475, and π* > 0.588 was the data value that determined LS was more efficiently solubilized in all the co-solvents/P6-Lev than in P6-Lev. Analogously, for all the PA-based PDESs mixed co-solvents with π* > the critical value, LS could be effectively dissolved, except for Cyrene/PA6-Lev with a low α value. In accordance with the LS dissolution mechanisms in the co-solvent/PDESs, the solubilization of EHL and OL also depended on the critical values of the co-solvents/PDESs (Fig. S9 and S10†). From the above results, we could conclude that lignin could be effectively solubilized by adding polar aprotic solvents (NFM, CP, DMI, and Cyrene), and the solubilization capability of the co-solvent could be visualized by π*. A new organic solvent solubilization mechanism in PDESs was proposed, whereby since π* was larger than the critical value, polar aprotic solvents as co-solvent showed excellent solubilization effect, which can be applied to various technical lignin samples (LS, EHL, and OL).

Regeneration of lignin from co-solvent/PDES solvent systems

The three types of lignin dissolved in the co-solvent/PDES solvent systems could be regenerated by employing water or absolute ethanol as anti-solvents. The characterization of native and regenerated lignin was initially evaluated using LS as a model (Fig. 7). The morphology changes of the native and regenerated lignin were observed by their scanning electron microscopy (SEM) images. The native LS exhibited a compact, ordered, and smooth surface structure, while the regenerated LS showed a relatively looser architecture (Fig. 7a and Fig. S11†).

Fourier-transform infrared (FT-IR) spectra were recorded to further probe the structural changes of the native LS and regenerated one by ethanol after the dissolution in 20 vol% DMI/P2-Lev solvent system (Fig. 7b). The FT-IR spectra of the native and regenerated LS obtained with ethanol as an anti-solvent were similar, indicating that DMI/P2-Lev was a non-derivatizing solvent as it maintained the lignin chemical structure during the dissolution and regeneration process of LS.[46] All the spectra showed O–H vibrations at 3430 cm^{-1},[47] and C–H asymmetric and symmetric vibrations of methyl and methylene groups at 2935 and 2840 cm^{-1}.[48] The absorbance bands at 1040 cm^{-1} could be attributed to the S=O stretching vibration of lignosulfonate.[48] The signals around 1123 and 1328 cm^{-1} were generated by syringaldehyde and S units.[49] All the spectra showed typical signals of aromatic skeleton vibrations (1604, 1509, and 1419 cm^{-1}).[49] These results showed that co-solvent/PDES pretreatment had no significant effect on the lignin aromatic skeletons, with these remaining in the original structure of lignin after pretreatment. For the other lignins (OL and EHL), the FT-IR spectra of the native and regenerated lignin were similar (Fig. S14†), retaining the basic aromatic skeletal and without the presence of new peaks. Nevertheless, the spectrum of the regenerated EHL using ethanol as an anti-solvent was shifted (Fig. S14†), indicating that the intramolecular interaction of EHL could be weakened in the EHL dissolution and regeneration process.

Thermal gravimetric analysis (TGA) was conducted to evaluate the thermal stability of native and regenerated EHL (Fig. 7c). It was also found that the regenerated LS exhibited a higher devolatilization compared to native LS. The pyrolysis of lignin was divided into three mass loss stages in response to temperature variations. In the first stage, the weight loss occurring near 130 °C could be ascribed to the evaporation of retained moisture in the lignin samples.[47] The second stage, which appeared in the range of 200–500 °C, was the maximum degradation rate temperature of LS, which could be attributed to the fragmentation of the inter-unit linkages and the release of monomeric phenols during the decomposition of the lignin.[48,50] Finally, the organic matter with strong chemical bonds continued to be degraded during the pyrolysis of LS,[51] while the other regenerated and native lignin samples (OL and EHL) still remained unvolatilized in the third stage after heating to 700 °C or 800 °C (Fig. S15†), attributed to the formation of highly condensed aromatic structures.[52] Meanwhile, the TGA analysis results indicated that the thermostability of regenerated OL using water as an anti-solvent was slightly lower than the native OL (Fig. S15†), which may be due to differences in the noncovalent interactions.[46,53] On the contrary, the regenerated EHL was more stable than the native

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Table 1 Chemical compositions of original and pretreated sugarcane bagasse, residue yield, lignin yield, and delignification at various conditions

Sample[a]	Pretreatment condition	Residue yield (wt%)	Composition of residue[b] (wt%)				Delignification[e] (%)	Lignin yield[e] (%)
			Cellulose	Hemicellulose	AIL[c]	ASL[d]		
Sugarcane bagasse	—	100	40.14	25.85	18.77	0.99	—	—
$R_{NFM/P2\text{-}Lev}$	120 °C /1 h	89.91	44.57	25.03	17.15	0.80	18.33	7.72
$R_{Cyrene/P2\text{-}Lev}$	120 °C /1 h	89.22	44.95	25.21	17.13	0.83	18.91	7.73
$R_{PC/P2\text{-}Lev}$	120 °C /1 h	88.43	44.24	24.42	16.93	0.82	20.57	9.16
$R_{DMI/P2\text{-}Lev}$	120 °C /1 h	87.81	45.56	24.16	16.57	0.86	22.54	11.80
$R_{NFM/P2\text{-}Lev/H}$	75 mM H_2SO_4/120 °C /1 h	75.68	49.48	21.16	14.57	0.87	40.86	22.38
$R_{Cyrene/P2\text{-}Lev/H}$	75 mM H_2SO_4/120 °C /1 h	60.27	59.87	15.83	12.23	0.85	60.10	28.98
$R_{PC/P2\text{-}Lev/H}$	75 mM H_2SO_4/120 °C /1 h	53.55	64.74	11.01	8.33	0.78	75.31	39.24
$R_{DMI/P2\text{-}Lev/H}$	75 mM H_2SO_4/120 °C /1 h	50.40	65.86	10.24	6.72	0.80	80.82	43.85

[a] The volume ratio of co-solvent to PDES was 2 : 8. [b] The compositions of samples were analyzed according to the NREL method.[55] [c] AIL, acid insoluble lignin. [d] ASL, acid soluble lignin. [e] Based on the original lignin in raw sugarcane bagasse (%, w/w).

EHL (Fig. S15†), which may be due to the formation of the condensed linkages of the regenerated EHL.[54]

Compositional analysis of raw and pretreated sugarcane bagasse

Efficient fractionation of lignocellulose remains challenging. As shown in the 2D HSQC NMR spectra (Fig. S3†), three different lignin samples were derived from herbaceous plants, and P2-Lev showed excellent lignin dissolving performance in this work. Therefore, P2-Lev and four kinds of co-solvents (Cyrene, NFM, DMI, and PC) were applied in lignin extraction from sugarcane bagasse.

The obtained solid residues (denoted as, $R_{NFM/P2\text{-}Lev}$, $R_{Cyrene/P2\text{-}Lev}$, $R_{PC/P2\text{-}Lev}$, and $R_{DMI/P2\text{-}Lev}$) were obtained from sugarcane bagasse under treatment with different co-solvents/P2-Lev at 120 °C for 1 h. The compositional analysis results of the original and pretreated sugarcane bagasse are shown in Table 1. The untreated sugarcane bagasse was mainly composed of cellulose (40.14 wt%), hemicellulose (25.85 wt%), and lignin (19.76 wt%). As shown in Table 1, when the pretreatment was conducted in 20 vol% co-solvent/P2-Lev for 1 h, all the co-solvent/P2-Lev pretreatments reduced the contents of lignin to varied degree, and the lignin from sugarcane bagasse could be removed and the delignification efficiency was in the range of 18%–22%. From the previous solubility tests, 20 vol% DMI/P2-Lev played a key role in the dissolution of EHL and LS, which also affected the pretreatment of sugarcane bagasse. After 20 vol% DMI/P2-Lev treatment, the residue yield dropped to 87.81 wt%, and the delignification efficiency was the maximum (22.51%). Hence, 20 vol% DMI/P2-Lev was the best choice for the fractionation of lignocellulose.

Lignocellulosic fractionation was enhanced by the co-solvent, in which dilute acid was needed to cleave the ether bonds in lignin and among the lignin–carbohydrate complexes.[26,56] Therefore, 75 mM H_2SO_4 as a catalyst was added to the 20 vol% co-solvent/P2-Lev, and the obtained solid residues (denoted as, $R_{NFM/P2\text{-}Lev/H}$, $R_{Cyrene/P2\text{-}Lev/H}$, $R_{PC/P2\text{-}Lev/H}$, and $R_{DMI/P2\text{-}Lev/H}$) were obtained from sugarcane bagasse under treatment with different co-solvents/P2-Lev and 75 mM H_2SO_4 at 120 °C for 1 h. Lignin from sugarcane bagasse could be effectively removed, and the solid recovery and the delignification were in the range of 50.4%–75.68% and 40.86%–80.82%, respectively (Table 1). Similarly, DMI among the four co-solvents also demonstrated excellent delignification performance. These results indicated that the co-solvent/PDESs contributed to the fractionation of lignocellulose.

Conclusions

In conclusion, novel sustainable polar aprotic/PDESs were successfully prepared and applied for dissolving different types of lignin, including EHL, LS, and OL at room temperature. It was found that the lignin solubility varied with the type of co-solvents and PDESs. First, there was no correlation between the lignin solubility and the degree of polymerization of PEG oligomers in PDESs, while the influence of the terminal group of these chains could not be ignored. Specifically, the P-based PDESs showed superior performance for lignin dissolution than the PA-based PDESs. Meanwhile, NFM, PC, and DMI actively participated in the EHL dissolution process in the PDESs, with PC showing excellent EHL solubilization in the PA6-PDESs. For LS, all the co-solvents were capable of assisting in the dissolution of LS in the PDESs (except NFM, and DMI in P6-Lev), among which PC was more suitable in the P2- and P4-based PDESs, and Cyrene was more suitable for the P6-PDESs. For OL, all the polar aprotic solvents (except Cyrene/P2-Lev) played an obvious role in the solubilization effect of OL; in particular, the strong solvating capacity of DMI as a co-solvent was more efficient in dissolving OL in all the PDESs. We believe our PDES/co-solvents systems could be applied in several areas, including 3D printing, film packaging, and gel materials.

A new unified solubilization mechanism was proposed by the Kamlet–Taft parameters. Co-solvent/PDESs with π^* > the critical values showed excellent performance for the different lignin samples (EHL, OL, and LS) due to the dispersion forces of lignin and the interactions with the aromatic parts of lignin. The critical values depend on the degree of polymerization of the solvent and the type of lignin. All-green co-solvent/PDES solvent systems will not only contribute to the development of a new sustainable

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solvent system for technical lignin, but also allow access to the high-valued utilization of industrial biowastes.

Experimental

Materials

Enzymatically hydrolyzed lignin (EHL) and ethanol organosolv lignin (OL) were purchased from Shanghai Yunzhe New Material Technology Co., Ltd. Sodium lignosulfonate (LS), levulinic acid (Lev), poly(ethylene glycol) diacrylate 200 (PA2), poly(ethylene glycol) diacrylate 400 (PA4), poly(ethylene glycol) diacrylate 600 (PA6), polyethylene glycol 200 (P2), polyethylene glycol 400 (P4), and polyethylene glycol 600 (P6) were obtained from Shanghai Macklin Biochemical Co., Ltd. *N*-Formylmorpholine (NFM), and propylene carbonate (PC), dimethylisosorbide (DMI), and dihydrolevoglucosenone (Cyrene) were purchased from Beijing Innochem Technology Co., Ltd. Cyrene is a trademark of Circa. The molecular weight, impurities, and element analysis of different lignin samples are described in the ESI.†

Preparation of the all-green polar aprotic solvents-based solvent systems

The PDESs were synthesized based on a previous report with some modifications.[57,58] First, the desired HBAs (P2, P4, P6, PA2, PA4, and PA6) and levulinic acid (LA) were mixed at a molar ratio of 2 : 1, 1 : 1, or 1 : 2. Second, the mixture was stirred at room temperature on a magnetic stirrer for 30 min. Finally, the desired PDESs were obtained after the above liquid was heated at 60 °C for 24 h under vacuum to remove the contained water. Subsequently, the all-green solvents were prepared by mixing the PDESs with different volume fractions of polar aprotic solvents and then stirring for 30 min.

Solubility of lignin

According to the method of measuring the solubility of lignin in the literature,[6] we measured the solubility of lignin. Typically, 0.17 g lignin (m_{lig}) was added to about 1 mL of PDESs with different co-solvents, and the mixture was stirred at room temperature in a 600 rpm magnetic agitator for 3 h. After the reaction, the mixture was centrifuged in a 14 800 rpm centrifuge for 5 min to obtain the lignin insoluble, which was washed 3 times with deionized water or anhydrous ethanol to ensure complete solvent removal, and placed in a vacuum drying oven at 40 °C for 24 h at mass (m_f). The solubility of lignin was calculated according to the following eqn (1):

$$\text{Solubility}\,(\%) = \frac{m_{lig} - m_f}{m_{lig}} \times 100\% \quad (1)$$

Measurement of the solvatochromic parameters

There are many dyes as probe molecules that can be used to determine solvatochromic parameters, and they do not all report identical values.[59] Here, we used 4-nitroaniline dye and Nile red to determine the solvatochromic parameters according to the method of measuring the solvatochromic parameters in the literature.[5]

The dyes were first dissolved in methanol with a concentration of 1.0×10^{-3} mol L^{-1}. Each time, 50 μL dye solution was added into a centrifugal tube. The residual methanol was then carefully removed by vacuum drying at 40 °C for 30 min. After removing the methanol, the dye was mixed with 2 g solution for detection. The dye/co-solvent/PDESs were placed into quartz cells. The absorption spectra were recorded using a Varian Cary 50 UV–visible spectrophotometer at 25 °C. The solvatochromic parameters were calculated according to the following equations:[60]

$$V_{NR} = \frac{1}{\lambda_{(NH_2)\,max}} \times 100\% \quad (2)$$

$$\alpha = \frac{19.9657 - 1.0241 \times \pi^* - V_{NR}}{1.6078} \quad (3)$$

where $\lambda_{(NR)max}$ is the wavelength corresponding to the maximum absorption of Nile red:

$$\pi^* = \frac{19.839 - V_{NR}}{2.9912} \quad (4)$$

$$\beta = 11.134 - \frac{3580}{\lambda_{(NH_2)\,max}} - 1.125 \times \pi^* \quad (5)$$

where $\lambda_{(NH_2)max}$ is the wavelength corresponding to the maximum absorption of 4-nitroaniline.

Pretreatment of sugarcane bagasse with co-solvent/PDESs solvent systems

The pretreatment of sugarcane bagasse was carried out according to the reported route with slight changes,[57] and the detailed steps were as follows:

Typically, 1 g sugarcane bagasse was mixed carefully with 10 g 20 vol% co-solvent (Cyrene, NFM, DMI, and PC)/P2-Lev in the capped reaction vessels. The lignocellulose and co-solvent/P2-Lev were heated in an oil bath at 120 °C with stirring for 1 h. After fractionation, the mixture was cooled to room temperature. Then, 20 mL anhydrous ethanol was added to the reaction vessel to dilute the viscous mixture, and the diluted mixture was then separated by vacuum filtration and washed with deionized water to remove the co-solvent/P2-Lev until the filtrate was neutral. Then the dissociated lignin was regenerated by adding deionized water to the filtrate. The cellulose-rich solid residues were dried for further composition analysis. The obtained solid residues were labeled as, for example, $R_{NFM/P2\text{-}Lev}$, $R_{Cyrene/P2\text{-}Lev}$, $R_{PC/P2\text{-}Lev}$, and $R_{DMI/P2\text{-}Lev}$, depending on the solvent system. All the processes were performed at least in duplicate. The residue yield (wt%), delignification (%), and lignin yield (%) were calculated as follows:

$$\text{Residue yield}\,(\text{wt}\%) = \frac{m_1}{m_2} \times 100\% \quad (6)$$

$$\text{Lignin yield}\,(\%) = \frac{m_3}{m_4} \times 100\% \quad (7)$$

$$\text{Delignification}\,(\%) = \left(1 - \frac{m_5}{m_4}\right) \times 100\% \quad (8)$$

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where m_1 is the weight of dried cellulose-rich solid residues, m_2 is the weight of raw sugarcane bagasse, m_3 is the weight of recovered lignin, m_4 is the weight of the total (acid soluble and acid insoluble) lignin in raw sugarcane bagasse, and m_5 is the weight of the total lignin remaining in the pretreated solid residues.

Also, 75 mM sulfuric acid was added to the reaction bottle as the experimental group, while all the other operations were the same as the above steps.

Characterizations

The viscosity and density of the prepared PDESs at room temperature were measured by an Anton Paar DMA 5000 M density meter and analysis of the phase-transition temperatures of the PDESs was conducted on a TA Discovery DSC 250 system, under the temperature ranges from −80 °C to 25 °C with a heating rate of 5 °C min^{-1}. Meanwhile, Fourier-transform infrared (FT-IR) spectra of lignin were recorded on a Bruker Tensor II spectrometer. Dried samples were ground and pelletized with KBr, and their spectra were recorded from 400 to 4000 cm^{-1} at a resolution of 4 cm^{-1} and 32 scans per sample. The TGA spectra of lignin were carried out using a TGA5500 system, from 30 °C to 800 °C at a rate of 10 °C min^{-1} in a nitrogen atmosphere. Scanning electron microscopy (SEM) images were collected on a HITACHI UHR FE-SEM SU8010 instrument (HITACHI, Tokyo, Japan). Steady shear and dynamic rheological measurements were conducted on an Anton Paar GmbH rheometer (the solubility of EHL in 20 vol% DMI/P2-Lev was calculated on the basis of 15 g EHL). The weight-average (M_w) and number-average (M_n) of the EHL and OL samples were obtained by gel permeation chromatography (GPC) (Agilent 1200) with a mobile phase of chromatographically pure THF at a flow rate of 1.0 mL min^{-1}. Monodisperse polystyrene was used for calibration of the molecular weight. The relative molecular weights of LS were measured by GPC (Waters 1515), using polystyrene sulfonate as the standard. The mobile phase was $NaNO_3$ solution (pH 11) for LS with a 0.5 mL min^{-1} flow rate, and the column temperature was 30 °C. Element analysis of C, H, N, O, and S was carried out on an organic element analyzer (Thermo FLASH2000). The sugar content (%, w/w), purity (%) and molecular weight (g mol^{-1}) of various lignin samples were determined by a high-performance anion-exchange chromatography system (Agilent, 1260) equipped with a Bio-Rad HPX-87P ion-exclusion column. Degassed 0.01 N dilute H_2SO_4 solution was used as the mobile phase with a flow rate of 0.6 mL min^{-1}, while the temperature in the column was kept constant at 50 °C. Two-dimensional heteronuclear single quantum coherence (2D HSQC) NMR spectra of the various lignin samples were obtained on a Bruker Avance 400 MHz (Bruker, Germany) spectrometer.

Author contributions

Qiaoling Liu: conceptualization, investigation, methodology, and writing – original draft. Yang Wang: investigation and validation. Jun-Li Ren: investigation. Jing Bian: writing – review & editing. Ming-Fei Li: conceptualization, validation, and writing – review & editing. Xiang Hao and Feng Peng: supervision, writing – review & editing, and funding acquisition.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This research was funded by the National Key Research and Development Project of China (2022YFB4201904), National Natural Science Foundation of China (22208022), National Science Fund for Distinguished Young Scholars of China (32225034), Beijing Forestry University Outstanding Young Talent Cultivation Project (2019JQ03017), and Ministry of Education, China-111 Project (BP0820033).

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